Chemotherapy for testicular cancer

Some side effects of treatment for testicular cancer may take a long time to improve. Sometimes they can become permanent, or develop years after treatment.

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What is chemotherapy?

Your doctor may suggest having chemotherapy treatment. Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy cancer cells. The drugs travel around the bloodstream, destroying possible cancer cells anywhere in the body.

You may have chemotherapy:

- after surgery, to reduce the risk of testicular cancer coming back (adjuvant chemotherapy)
- to treat testicular cancer that has spread outside the testicle, or come back after an orchidectomy
- to treat testicular cancer that has come back after initial chemotherapy.

The drugs most commonly used to treat testicular cancer are:

- bleomycin
- etoposide
- cisplatin.

This combination is called BEP chemotherapy.

You may also have a chemotherapy drug called carboplatin.

We have more detailed information about chemotherapy.

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Adjuvant chemotherapy

You may have adjuvant chemotherapy to reduce the risk of testicular cancer coming back.

- If you have an early (stage 1) non-seminoma, you may have 1 cycle of BEP. Each cycle is 3 weeks.
- If you have an early (stage 1) seminoma, you may have a single treatment with carboplatin.

If the cancer has spread or comes back during monitoring (surveillance)

The chemotherapy drugs you have will depend on the stage of disease at diagnosis and previous cancer treatment. You may have 3 or 4 cycles of BEP if testicular cancer has:

- spread outside the testicle
- come back during monitoring (surveillance).

Sometimes, bleomycin is left out. You may have 3 or 4 cycles of etoposide and cisplatin (EP) instead. Your oncology doctor will explain which is the best chemotherapy combination for you and how many treatments you may have.

Bleomycin may cause breathing problems if you smoke or already have a lung condition.

Intensive chemotherapy

If BEP does not get rid of the cancer, or the cancer comes back, you will usually have more intensive chemotherapy. Depending on the stage of the cancer, you may have intensive chemotherapy straight away.

Some drug combinations are:

- PEI (cisplatin, etoposide, ifosfamide) – this drug combination is also sometimes called VIP
- TIP (paclitaxel, ifosfamide, cisplatin)
- VeIP (vinblastine, ifosfamide, cisplatin).

High-dose chemotherapy with stem cell support (autologous stem cell transplant)

Occasionally, high-dose chemotherapy with stem cell support may be used for testicular cancer. You may have this as part of a clinical trial.

Stem cells produce blood cells. The high-dose treatment destroys stem cells in the bone marrow, as well as the cancer cells. Because of this, doctors remove some stem cells before treatment. These are then stored to give back to you after treatment.

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The stem cells are given back through a drip (infusion). This is called an autologous stem cell transplant. The cells go to your bone marrow and start to make blood cells again.

This type of treatment allows you to have higher than usual doses of chemotherapy. You usually have it if the cancer has not responded completely to previous treatment. Or you may have it if the cancer has come back after treatment. It is used to increase the chances of curing the cancer.

You may have high-dose treatment in hospitals with large cancer units that specialise in this type of treatment. This is an intensive treatment, and you will need to stay in hospital for several weeks.

How chemotherapy is given

You usually have chemotherapy as an outpatient. You have it as an injection or through a drip. This is called intravenous chemotherapy. You may have the drugs through a:

- cannula – a short, thin tube the nurse puts into a vein in your arm or hand
- central line – a fine tube that goes under the skin of your chest and into a vein close by
- PICC line – a fine tube that is put into a vein in your arm and goes up into a vein in your chest.

You have chemotherapy in cycles of treatment. A cycle usually takes 3 weeks.

If you are having BEP, you usually have to go to the hospital for 3 to 5 days as an outpatient. Sometimes you stay overnight. This is followed by weekly outpatient visits to have the rest of your cycle of treatment.

Other combinations of chemotherapy drugs might mean you spend more time in hospital and have the drugs more often. Your doctor or nurse will tell you more about your treatment options, how many cycles of treatment you need, and how you will have it.

Side effects of chemotherapy for testicular cancer

Chemotherapy drugs causes side effects. But there are usually ways these can be controlled. The side effects can depend on the type of chemotherapy drugs used. Your doctor or nurse will give you information about the likely side effects of your treatment.

We have more information about the side effects of chemotherapy.

Fertility and storing sperm before treatment

Your doctor will advise you to consider storing sperm (sperm banking) before having treatment. This is because chemotherapy for testicular cancer usually causes infertility during treatment and for a time after. This is usually temporary, and your fertility will usually recover after treatment. How quickly your sperm count recovers varies from person to person. It generally starts to recover about 18 to 24 months after treatment. But it can take up to 5 years to recover. If you have a low sperm count before chemotherapy treatment, sperm production can sometimes improve after treatment.

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If you are having high-dose chemotherapy, you have a much higher risk of infertility. And infertility caused by this kind of treatment is usually permanent. If you are concerned about this, you can talk to a specialist.

Contraception

Although treatment may affect your fertility, it is not always possible to know when this will happen. You may still be able to get someone pregnant. Your doctor will advise you not to make someone pregnant during treatment. This is because the chemotherapy drugs can temporarily damage your sperm, and possibly harm a developing baby.

It is usually best to use a barrier method of contraception, such as a condom, while you are having treatment.

It is also important to continue using effective contraception for about a year after chemotherapy. This allows time for your sperm to recover from any damage that treatment may have caused. You can talk to your doctor or nurse about this.

About our information

References

Below is a sample of the sources used in our testicular cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

The British Association of Urological Surgeons (BAUS) Guidelines on Testicular Cancer. March 2015. Available from: https://www.baus.org.uk/professionals/sections/testicular_cancer.aspx (accessed April 2022).

European Association of Urology (EAU) Guidelines on Testicular Cancer 2022. Available from: https://d56bochluxqnz.cloudfront.net/documents/full-guideline/EAU-Guidelines-on-Testicular-Cancer-2022.pdf (accessed April 2022).

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Reviewers

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, Dr Ursula McGovern, Consultant Medical Oncologist.

Our cancer information has been awarded the PIF TICK. Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

Date reviewed

Reviewed: 01 May 2022 | Next review: 01 May 2025

This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

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How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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